Unit III
Indian Science and Technology

Content:

3.1 History of Science in India,
3.2 India's contributions to the world of science,
3.3 Developments in astronomy,
3.4 Developments in mathematics
3.5 Developments in chemistry
3.6 History of Technology in India
- 3.6.1 Agriculture
- 3.6.2 Early Craft Techniques
- 3.6.3 Textiles
- 3.6.4 Paper and Writing
- 3.6.5 Water Management System.
- 3.6.6 Metallurgy
- 3.6.7 Constructions

3.1 History of Science in India:

- **Science and mathematics** were highly developed in India during the ancient period.
- Baudhayan, Aryabhatta, Brahmgupta, Bhaskaracharya, and Mahaviracharya were some famous ancient Indian mathematicians.
- Kanad, Varahamihira, and Nagarjuna were some well-known scientists.
- **Medical science** was also advanced in ancient India.
- **Ganita** is the general name for mathematics, which includes Arithmetic, Geometry, Algebra, Astronomy, and Astrology.
- **Arithmetic** is known by several names, including Pattin Ganita (board calculations) and Anka Ganita (calculations with numerals).
- **Geometry** is known as Rekha Ganita (line works), and Algebra, Bija Ganita (seed analysis), Astronomy, and Astrology are all included in the term Jyotisa.
- India has a long history of science and technology.

3.2 India's contributions to the world of science

1. Idea of Zero
2. Decimal System
3. Numerical Notations
4. Binary Numbers
5. Chakravala Method of Algorithms
6. Ruler Measurements

7. Ayurveda

1. Idea of Zero

- **Aryabhata,** a mathematician, was the first to create a symbol for zero, and it was through his efforts that mathematical operations such as **addition** and **subtraction** began to use the digit zero.
- The concept of zero and its incorporation into the **place-value system** also allowed for the writing of numbers of any size using only ten symbols.

2. Decimal System

- India invented the decimal system, which uses ten symbols to represent all numbers.
- In this system, each symbol was assigned a **positional value** as well as an **absolute value.**
- This system made the use of arithmetic in practical inventions much faster and easier due to the simplicity of the decimal notation, which facilitated calculation.

3. Numerical Notations

- India developed a system of different symbols for each number from one to nine as early as 500 BCE.
- The Arabs, who called it the hind numerals, adopted this notation system. Centuries later, the western world adopted this notation system, dubbed Arabic numerals because it arrived via Arab traders.

4. Binary Numbers

- The basic language in which computer programs are written is binary numbers.
- Binary is essentially a set of two numbers, 1 and 0, the combinations of which are known as bits and bytes.
- **Pingala,** a Vedic scholar, first described the binary number system in his book **Chandahsastra.**

5. Chakravala Method of Algorithms

- The chakravala method is a cyclic algorithm for solving indeterminate quadratic equations such as Pell's equation.
- **Brahmagupta,** a well-known mathematician of the 7th century CE, developed this method for obtaining integer solutions.
- **Jayadeva,** another mathematician, later generalised this method for a broader range of equations, which was refined further by Bhskara II in his Bijaganita treatise.

6. Ruler Measurements

- **Excavations at Harappan sites** have yielded ivory and shell rulers or linear measures.
- The calibrations, which are marked out in minute subdivisions with amazing accuracy, correspond closely with the hasta increments of 1 3/8 inches, which were traditionally used in South Indian ancient architecture.
- The dimensions of ancient bricks discovered at excavation sites correspond to the units on these rulers.

7. Ayurveda

- Charaka, known as the Father of Indian Medicine, was the first physician to introduce the concepts of digestion, metabolism, and immunity in his book.
- For two millennia, Charaka's ancient manual on preventive medicine remained a standard work on the subject, and it was translated into many foreign languages, including Arabic and Latin.

3.3 Developments in Astronomy

- Aryabhatta's theories marked a significant departure from astrology, which emphasized beliefs over scientific explorations.
- In ancient India, astrology and horoscopes were studied.
- Astronomy has made great strides. **Planetary motion** became emphasized and closely monitored.
- **The Jyotishvedanga** texts established systematic categories in astronomy, but Aryabhatta(499 AD) dealt with the more fundamental issue.
- **Aryabhatta** deviated from Vedic astronomy and gave it a scientific perspective, which later astronomers followed.

3.4 Mathematics developments in mathematics and chemistry

- **Harappa's town planning** demonstrates that the people were well-versed in measurement and geometry.
- **The notation system**, **the decimal system**, and the **use of zero** were the three most important contributions to mathematics.
- The Arabs brought the notations and numerals to the West. These numerals took the place of Roman numerals. In the second century BC, Zero was discovered in India.
- **Brahmagupta's Brahmasputa Siddhanta** was the first book to mention 'zero' as a number; thus, Brahmagupta is known as the man who discovered zero. He explained how to use zero with other numbers.
- **Aryabhatta** discovered algebra as well as the area of a triangle, which led to the development of trignometry.
- Mathematics and astronomy, when combined, sparked an interest in time and cosmology.
- These astronomical and mathematical discoveries became the foundations for further research and progress.

3.5 Developments in chemistry

- Chemical techniques in India can be traced back all the way to the Indus valley or Harappan civilisation (3rd millennium BCE).
- Following Acharya Prafulla Chandra Ray (1861-1944), the eminent Indian chemist of the last century and a historian of chemistry, five stages in its development in India can be recognised. They are:

(i) The pre-Vedic stage upto 1500 BCE, including the Harappan period,
(ii) The Vedic and the Ayurvedic period upto 700 CE,
(iii) The transitional period from 700 CE to 1100 CE,
(iv) The Tantric period from 700 CE to 1300 CE, and
(v) The 'Iatro-Chemical period' from 1300 CE to 1600 CE.

3.6 History of Technology in India

3.5.1 Agriculture
3.5.2 Early Craft Techniques
3.5.3 Textiles
3.5.4 Paper and Writing
3.5.5 Water Management System.
3.5.6 Metallurgy
3.5.7 Constructions

3.6.1 Agriculture

- Barley and wheat cultivation—along with the rearing of cattle, sheep and goat—was visible in Mehrgarh by 8000-6000 BCE.

3.6.2 Early Craft Techniques

- Early humans developed technologies such as **stone-working**, **agriculture, animal husbandry**, **pottery**, **metallurgy**, **textile manufacture, wood-carving**, **sailing**, and so on with little science to back them up.

3.6.3 Textiles

- For about 3,000 years from 1500 BC to 1500 AD, India had the monopoly in manufacturing cotton textiles.
- India has a rich regional textile tradition.
- That can be traced as far back as the Indus Valley Civilization. Records of ancient Indian textile traditions exist mostly in literature and in sculptures.

3.6.4 Paper and Writing

- In India, the available writing materials were generally of two types: hard and soft.
- Stone, metal, shells and earthenware were the examples of hard material.
- Engraving, embossing, painting and scratching were used for writing.

3.6.5 Water Management System.

- The Harappan civilization epitomizes the level of development of water sciences in ancient India that includes construction of **sophisticated hydraulic structures**, **wastewater disposal systems** based on centralized and decentralized concepts, and methods for **wastewater treatment**.

3.6.6 Metallurgy

- The first form of **crucible steel was wootz**, developed in India some time around 300 BCE. (In its production the iron was mixed with glass and then slowly heated and then cooled. As the mixture cooled the glass would bond to impurities in the steel and then float to the surface, leaving the steel considerably purer.)
- **Glazed potteries** and bronze and copper artefacts (an object that is made by a person) discovered in the **Indus Valley excavations i**ndicate a highly developed metallurgy.
- By the first century AD, **mass production of metals** such as iron, copper, silver, and gold, as well as alloys such as brass and bronze, was underway.
- The **iron pillar** in the **Qutub Minar** complex demonstrates the high quality of alloying that took place.
- **Dyeing textiles** was popular. The quality of colour is reflected in the Ajanta frescoes. These paintings have survived to the present day.

3.6.7 Constructions

- India has been a pioneer in the field of architecture since the Indus Valley Civilization. The Indus urban system serves as an inspiration for modern cities.
- Buildings, pillars, cave construction, and chaitya construction were examples of advanced architecture in India during the Mahajanapada and Maurya periods.
- Temples were advanced in ancient India. The Kailashnath temple, built on a hillside, is a marvel of engineering.